Query Doctor

Same rows. Faster. Or the doctor *won't sign.*

Executes the original and the rewrite on Zetaris; signs only when the rows match. Hero: **2.11×** over 10 paired runs.

Case report hero-r1 — ACCEPTED

SPECIMEN · THE REWRITE DROPS THE STRUCK LINE

```
SELECT COUNT(1) AS n
FROM   TPCDS_DB.store_sales ss
~~LEFT JOIN TPCDS_DB.promotion p ON ss.ss_promo_sk = p.p_promo_sk~~
WHERE  ss.ss_customer_sk IS NULL
```

RESULTS VS REFERENCE · PER-SIDE MEDIANS OF 3 ALTERNATING PAIRS



original	87cd00c5
rewrite	87cd00c5

ROW-IDENTICAL · 87CD00C5

Verified by **Verifier** — code, no model · 2.1× = median pair ratio, 3/3 pairs faster

landing + this deck
querydoc-flax.vercel.app

PROBLEM · WHO SIGNS OFF

Faster can still be *wrong*.

```
  SELECT COUNT(1) AS n
  FROM   TPCDS_DB.store_sales ss
- LEFT JOIN TPCDS_DB.promotion p
-   ON ss.ss_item_sk = p.p_item_sk
  WHERE  ss.ss_customer_sk IS NULL
```

original count	129,776
after the edit	129,752

Reviewers see one deleted join. Users: whoever reviews a rewrite before it merges — the diff is all they see.

SOLUTION

A model proposes. *Code decides.*

CODE

Verifier

Executes both, compares rows, times pairs.

1 original runs twice — deterministic?

2 same columns and types

3 0 rows differ (or equal row hash)

4 ≥ 1.2×, saving ≥ 1,782.9 ms, most pairs won

MODEL

Diagnoser

Reads the plan and the join key.

`unused_outer_join, node 7`

MODEL

Rewriter

One rewrite per hypothesis.

`drops the unused join`

CODE

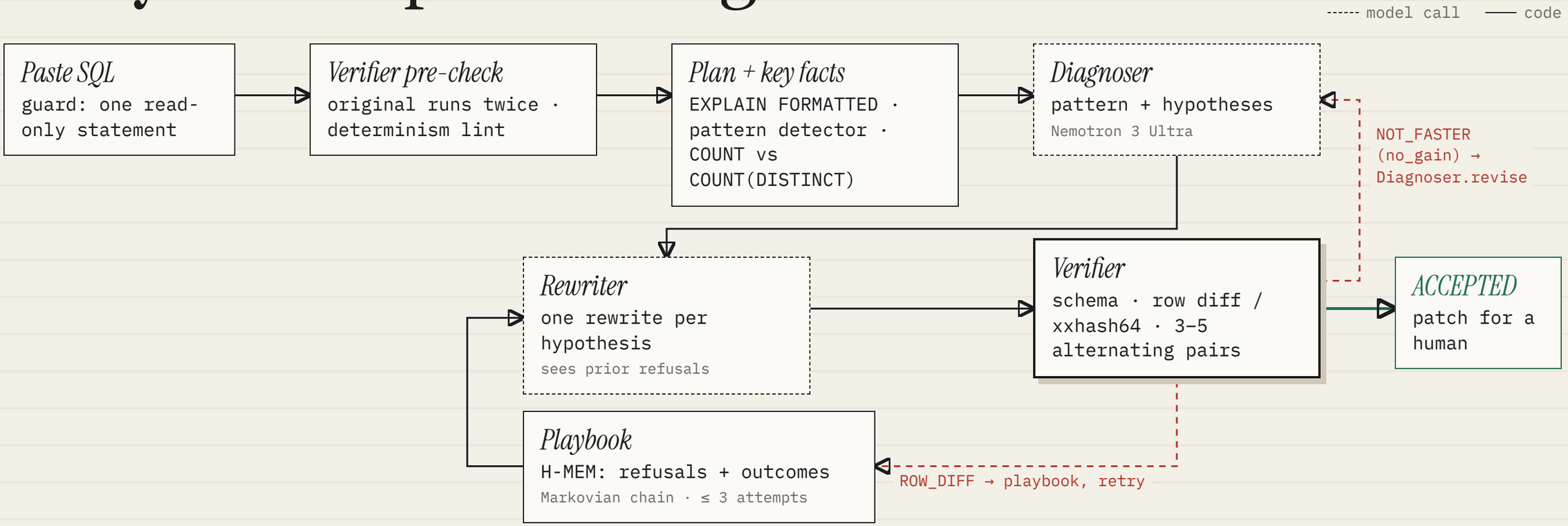
Orchestrator

Sequences, retries, remembers refusals.

`1 attempt → ACCEPTED`

WORKFLOW · ARCHITECTURE

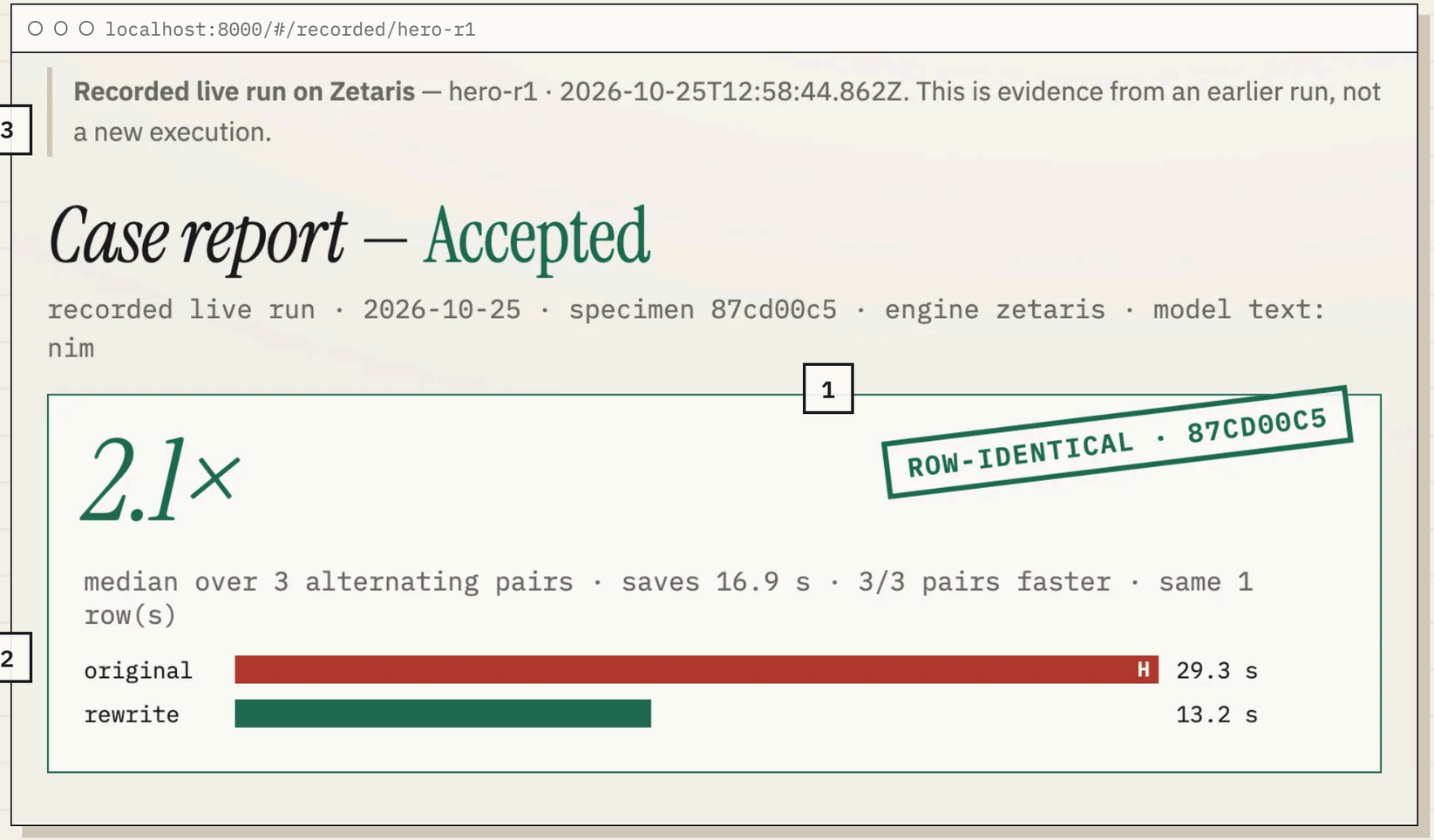
Every rewrite passes *one gate.*



Engines	Zetaris · local platform over a Java JDBC sidecar (live)	DuckDB · committed NYC TLC subset (sample, keyless)	model: Nightshift default · NIM fallback (recorded)

PRODUCT

The report a reviewer *signs*.



1. Verdict and stamp, first.
2. Measured bars: original vs rewrite.
3. Labelled: evidence from a recorded run.

THE MOMENT

One edit. Two keys. *Two verdicts.*

Case report hero-r1 ACCEPTED

```
LEFT JOIN TPCDS_DB.promotion p
  ON ss.ss_promo_sk = p.p_promo_sk
```

ON p_promo_sk **300 distinct** of 300 non-null

fingerprint 87cd00c5 = 87cd00c5
2.1× median pair ratio · 3/3 faster

ROW-IDENTICAL · 87CD00C5

Case report hero-trap-r1 REFUSED

```
LEFT JOIN TPCDS_DB.promotion p
  ON ss.ss_item_sk = p.p_item_sk
```

ON p_item_sk **291 distinct** of 293 non-null

attempt 1 fingerprint e51f9da7 ≠ 87cd00c5
first differing row 129752

REFUSED · ROW_DIFF

EVIDENCE

Measured on the engine, *not estimated.*

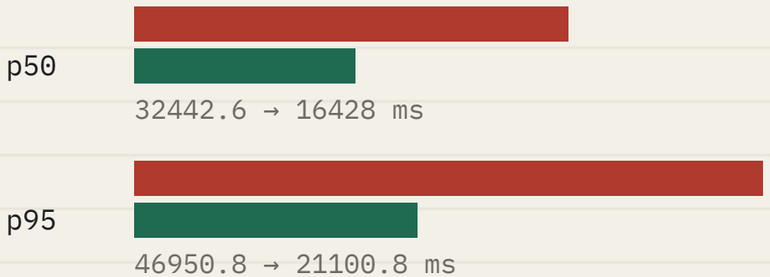
2.1×

live verdict · 3 pairs

hero-r1 · 3/3 pairs
hero-r2 · 2.15×
fingerprint 87cd00c5

2.11×

bench, 10 alternating pairs



red original · green rewrite

6 / 15

baseline changed rows

one model call, same Verifier, DuckDB
Query Doctor shipped: 0



INSTRUMENTS

Zetaris runs it. *Meterless remembers it.*

Zetaris

11 calls

9	Verifier runs
1	key profile
1	`EXPLAIN FORMATTED`

NVIDIA Nemotron · 2 calls

Meterless

3 calls

2	H-MEM open, query
1	Markovian chunk

Nightshift: 0 live runs, mock-tested

counts from the recorded trace of hero-r1 · Nightshift: 16 mock-runtime tests

STACK

Small, tested, *runs keyless.*

```
querydoc/
├─ src/agents/           diagnoser · rewriter · verifier · patterns · profile
├─ src/verify/           canonical · diff · fingerprint · timing · order · determinism
├─ src/reasoning/        chain.ts — Markovian
├─ src/memory/           playbook.ts — H-MEM
├─ src/adapters/         nightshift · nim · llm
├─ engine/zetaris-jdbc/  QuerySidecar.java — one warm JDBC connection
├─ vendor/meterless/     upstream engines, unmodified
├─ test/                 26 files
└─ Dockerfile            SAMPLE_MODE=true, no keys
```

286

tests, no keys needed

TypeScript · Node 22 · DuckDB sample engine

Cursor wrote scaffold d3e823c; Claude Code the rest.

WHAT WAS UNIQUE

It keeps the *wrong answers.*

ON THE PAGE

Refused rewrites stay, struck through, with the differing row.

IN MEMORY

A repeated refusal is read from the ledger.

IN THE REPORT

"identical on the current data" — not a proof.

1 hypothesis h1 · REJECT (ROW_DIFF)

Rewriter's note — model text ~~Removed unnecessary LEFT JOIN to promotion table since no promotion columns are referenced and the join cannot affect the row count of a LEFT JOIN with COUNT(1).~~

rows differ: 1 vs 1 — first differing row ["129752"] (rewrite only)

▸SQL — diff against the original

2 hypothesis h1 · REJECT (ROW_DIFF)

Rewriter's note — model text ~~Replace LEFT OUTER JOIN with LEFT SEMI JOIN to preserve row-count semantics while avoiding broadcast hash join overhead by eliminating duplicate matches from non-unique p_item_sk.~~

rows differ: 1 vs 1 — first differing row ["2853"] (rewrite only)

▸SQL — diff against the original

3 hypothesis h2 · REJECT (ROW_DIFF)

Rewriter's note — model text ~~Removed the redundant LEFT JOIN to promotion table since no promotion columns are used and the join cannot affect COUNT(1) result~~

refusal reused from ledger mem-11 — same original-result fingerprint; not re-run

▸SQL — diff against the original

FUTURE

Next: queries *we didn't write.*

- ☐ Queries not designed by us, baseline on Zetaris
- ☐ Re-record the hero through a live Nightshift runtime
- ☐ Zetaris MCP intake (`get_slow_queries`)
- ☐ Write-back of an approved rewrite

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Verified by the Verifier.